

● EASY

● ADVANCED MATH

● ANSWER KEY

Advanced Math

02

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ANSWER KEY · SAT QUESTION BANK

2026-05-29

Q1**C**

C. $p = 240\left(\frac{1}{2}\right)^r$

Choice C is correct. It's given that one of the two players in a match is eliminated during each round. Therefore, the number of players decreases by half at the end of each round, and so the number of players eliminated at the end of a round also decreases by half. A decreasing exponential equation can be written in the form $p = a(1 - b)^r$, where b is the value of the decrease each round, r , and a is the value of p , the number of players, when $r = 0$. It's given that there are 240 players in the tennis competition. Therefore, $a = 240$. Since the number of players eliminated at the end of a round decreases by half, it follows that $b = \frac{1}{2}$. Substituting 240 for a and $\frac{1}{2}$ for b in the equation $p = a(1 - b)^r$ yields $p = 240\left(1 - \frac{1}{2}\right)^r$, or $p = 240\left(\frac{1}{2}\right)^r$.

Choice A is incorrect. This equation gives the number of players eliminated at the end of round r for a competition in which there are 15, not 240, players.

Choice B is incorrect. This equation gives the number of players eliminated at the end of round r for a competition in which there are 15, not 240, players, and for which the number of players doubles, rather than decreases by half, after each round.

Choice D is incorrect. This equation gives the number of players eliminated at the end of round r for a competition in which the number of players doubles, rather than decreases by half, after each round.

Q2**C****C.** 1, 5

Choice C is correct. The solution to the system of two equations corresponds to the point where the graphs of the equations intersect. The graphs of the linear function and the absolute value function shown intersect at the point 1, 5. Thus, the solution to the system is 1, 5.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect. This is the y-intercept of the graph of the linear function.

Choice D is incorrect. This is the vertex of the graph of the absolute value function.

Q3**B****B.** $6x^3$

Choice B is correct. The given expression can be rewritten as $11x^3 + -5x^3$. Since the two terms of this expression are both constant multiples of x^3 , they are like terms and can, therefore, be combined through addition. Adding like terms in the expression $11x^3 + -5x^3$ yields $6x^3$.

Choice A is incorrect. This is equivalent to $11x^3 + 5x^3$, not $11x^3 - 5x^3$.

Choice C is incorrect. This is equivalent to $11x^6 - 5x^6$, not $11x^3 - 5x^3$.

Choice D is incorrect. This is equivalent to $11x^6 + 5x^6$, not $11x^3 - 5x^3$.

Q4**3**

The correct answer is 3. It's given that the y-intercept of the graph shown is 0, y. The graph passes through the point 0, 3. Therefore, the value of y is 3.

Q5**D**

D. $s = \frac{6r - t}{7}$

Choice D is correct. Subtracting t from both sides of the given equation yields $6r - t = 7s$. Dividing both sides of this equation by 7 yields $\frac{6r - t}{7} = s$. Therefore, the equation $s = \frac{6r - t}{7}$ correctly expresses s in terms of r and t .

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Q6**D**

D. $17x^2 - 10xy + 10y^2$

Choice D is correct. Expressions in the form $a^2 - b^2$ follow the difference of two squares pattern and can be factored as $(a - b)(a + b)$. In the given expression, $17(x^2 - 100y^2)$, the expression $x^2 - 100y^2$ follows the difference of two squares pattern. It follows that the expression $x^2 - 100y^2$ can be rewritten as $(x - 10y)(x + 10y)$. Therefore, the expression $17(x - 10y)(x + 10y)$ is equivalent to $17(x^2 - 100y^2)$.

Choice A is incorrect. This expression is equivalent to $17(x^2 - 52xy + 100y^2)$, not $17(x^2 - 100y^2)$.

Choice B is incorrect. This expression is equivalent to $17(x^2 + 48xy - 100y^2)$, not $17(x^2 - 100y^2)$.

Choice C is incorrect. This expression is equivalent to $17(x^2 - 20xy + 100y^2)$, not $17(x^2 - 100y^2)$.

Q7**3**

The correct answer is 3. The value of $f(0)$ is the value of y on the graph of $y = f(x)$ in the xy -plane that corresponds with $x = 0$. It's given that the value of $f(0)$ is an integer. For the graph of $y = f(x)$ shown, when $x = 0$, the corresponding integer value of y is 3. Therefore, the value of $f(0)$ is 3.

Q8**B****B.** -9

Choice B is correct. Since the point x, y is an intersection point of the graphs of the given equations in the xy -plane, the pair x, y should satisfy both equations, and thus is a solution of the given system. According to the first equation, $y = 76$. Substituting 76 in place of y in the second equation yields $x^2 - 5 = 76$. Adding 5 to both sides of this equation yields $x^2 = 81$. Taking the square root of both sides of this equation yields two solutions: $x = 9$ and $x = -9$. Of these two solutions, only -9 is given as a choice.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect. This is the value of coordinate y , rather than x , of the intersection point x, y .

Q9**B****B.** $6x^2$

Choice B is correct. Since each term in the given expression has a common factor of x^2 , it can be rewritten as $(13 - 7)x^2$, or $6x^2$. Therefore, the expression $6x^2$ is equivalent to $13x^2 - 7x^2$.

Alternate approach: Since the two terms of the given expression are both constant multiples of x^2 , they are like terms and can, therefore, be combined through subtraction. Subtracting like terms in the expression $13x^2 - 7x^2$ yields $6x^2$.

Choice A is incorrect. This expression is equivalent to $(13x)(-7x)$, not $13x^2 - 7x^2$.

Choice C is incorrect. This expression is equivalent to $13x^2 + 7x^2$, not $13x^2 - 7x^2$.

Choice D is incorrect and may result from conceptual or calculation errors.

Q10

B

B. $n = 53(2)^d$

Choice B is correct. It's given that each day for five days after a social media channel posted a video, the number of subscribers doubled from the number the previous day. Since the number of subscribers doubled each day, the relationship between n and d can be represented by an exponential equation of the form $n = ab^d$, where a is the number of subscribers at the time of posting the video and the number of subscribers to the channel increases by a factor of b each day. It's given that at the time of posting the video, the channel had 53 subscribers. Therefore, $a = 53$. It's also given that the number of subscribers doubled, or increased by a factor of 2, from the number the previous day. Therefore, $b = 2$. Substituting 53 for a and 2 for b in the equation $n = ab^d$ yields $n = 532^d$.

Choice A is incorrect. This equation gives the total number of subscribers to a channel for which the initial number of subscribers was 1 and the number increased each day by 53 times the number from the previous day.

Choice C is incorrect. This equation gives the total number of subscribers to a channel for which the number of subscribers each day was half the number from the previous day, rather than double the number.

Choice D is incorrect and may result from conceptual errors.

Notes

Accept equivalent forms (e.g. $0.25 \equiv 1/4$). For grid-in items, decimal and fraction answers are both valid.

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